Tiger Motors Lean Education Center
Student Built – Experimental Lean Learning

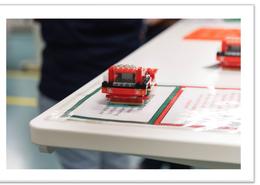
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Intro/Background

Auburn University Industrial & Systems Engineering Department launched Tiger Motors in 2011 enabling students to encounter an experiential learning environment simulating a high-volume automotive factory with LEGO® vehicles. Participants create a product from scratch while experiencing multiple systems working together and witnessing the transformation from a mass production environment to a lean production system. Giving students real-world experience implementing lean tools.

Lean Tools



One Piece Flow



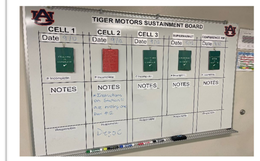
Heijunka Box



Kaizen Boards



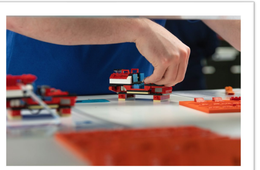
Standardization



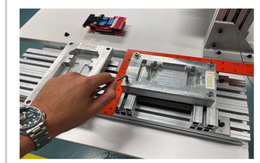
5S Sustainability System



JIT Material Flow



Jidoka



SMED



Kanban Cards

Objective

Provide a path for students to discover Lean's value for themselves. Activities in the lab can solidify an understanding of the concepts more strongly than a lecture or video. Lean is a culture and a way of thinking. The lab has a Team Structure, the Team Leader needs to feel the pressure to overproduce and the effect of one's natural tendencies to hoard parts and work ahead; before they can fully appreciate the Lean concepts of single piece flow and the importance of standardization.

"Data is of course important in manufacturing, but I place the greatest emphasis on facts"
– *Taiichi Ohno*

Going to the Gemba







Experimental Learning in Action-
Undergraduates, Graduates and Corporate partners

"Knowledge is something you buy with the money. Wisdom is something you acquire by doing it,"
– *Taiichi Ohno*



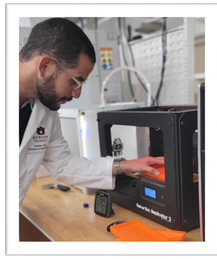
Speedster and SUV on the conveyor belt

"The gemba and the gembutsu have the information. We must listen to them."
- *Taiichi Ohno*

LEGO® Creator 3-in-1 kits are divided into 15 stations (one operator on each) and three cells (teams), four team leaders and three material handlers are in charge of their teams. The students' performance is measured by OEE.

Built and Maintained by Industrial and Systems Engineering Students

"Launching our own laboratory 10 years ago marked the beginning of a new and exciting phase in the ISE department at AU. But it was also an overwhelming challenge. Few of the students had any training or experience in the management skills necessary to build and run an effective lab. However, them having a mindset of looking for solutions has been helpful when encountering unexpected *disruptions that kept them motivated to develop and evolve the Tiger Motors lab."*
– *Current Lean GTA*



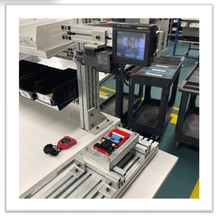


Students setting up the Tier 1 Sequence Supplier (War Eagle Inc.) manufacturing cell for demonstration

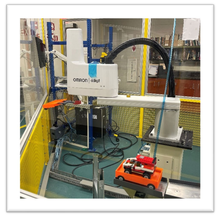
Technology



Light Guide Projection Based AR System

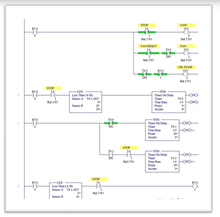


In-Station Quality Inspection Vision System

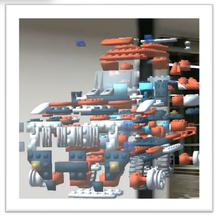


End Line Quality Sorting Robot

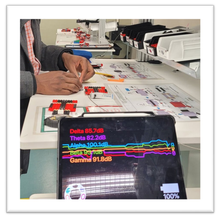
Research



Lean Systems and Industry 4.0 Interaction



5S Training with Augmented Reality



Neurodiversity worker assessment

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